

Grade 1 Ontario Curriculum Data

Data Detectives Take the Case

Unit Overview: Data Detectives Take the Case

Grade: Grade 1 | **Strand:** Data

Target Expectations: D1.1, D1.2, D1.3, D1.4, D1.5

Learning Goal: Children will sort by attribute, collect and record data with tallies, display data in pictographs, order frequencies, and draw conclusions from data displays.

Lesson	Lesson Title	Learning Goal
Day 1	Sort by One Rule	I can sort things by one rule and say what the rule is.
Day 2	Tally Marks Tell the Count	I can collect data and record it with tally marks.
Day 3	Pictograph Power	I can make a pictograph with pictures, a title, and labels.
Day 4	Order Greatest to Least	I can order categories from greatest to least.
Day 5	The Detective's Conclusion Capstone	I can ask and answer questions about a data display and write a conclusion.

Lesson 1: Sort by One Rule

Goal: Sorting groups things by ONE rule. The rule decides where each thing goes.

Expectation: D1.1

Learning Goal: "I can sort things by one rule and say what the rule is."

Materials: M1_sort_objects, M7_data_chart, M9_attribute_cards, M10_detective_poster

1. Minds On (12 mins): Detective Dot Sorts the Pile

Detective Dot: I have a PILE of stuff. Help me SORT it. What ONE rule can we use?

Bring children to the carpet. Detective Dot + Tally puppets. 'Friends — I'm Detective Dot the Sorting Squirrel, and this is my partner Tally the Tally Cat. Today we learn the FIRST detective skill: SORTING.'

2. Action (25 mins): Sorting Stations

- Whole group: sort 4 piles.
- Three stations.
- Stretch challenge at Station C: sort by 2 rules at once (red AND big).
- Discuss naming the rule.
- Circulate to record diagnostic evidence on the sorting tracker.

3. Consolidation (13 mins): Detective Dot's Sort Brag

Discuss:

- What is a RULE?
- Can the same pile be sorted in DIFFERENT ways?
- How do you SAY your rule?
- What happens if you change the rule mid-sort?
- STRATEGY COMPARE + LOOK AHEAD: Two pairs share different rules on the same pile — surface that the rule (not the objects)...

Exit: Whole class chants Detective Dot's refrain: "Pick the rule, SAY the rule, SORT the rule — case CLOSED!" Then each child writes ONE sort on a sticky: 'I sorted ___ by ___.'

Lesson 2: Tally Marks Tell the Count

Goal: We collect data by asking and counting. Tally marks group by FIVES so we can count fast.

Expectation: D1.2

Learning Goal: "I can collect data and record it with tally marks."

Materials: M2_tally_chart, M5_question_cards, M7_data_chart, M8_tally_chart

1. Minds On (12 mins): Tally Counts the Class

Tally: Who likes APPLES? Hands up! Now let's count with TALLY MARKS — one line per hand!

Bring children to the carpet. Detective Dot + Tally puppets. 'Friends — yesterday we sorted. Today we COLLECT DATA with TALLY MARKS.'

2. Action (25 mins): Survey Stations

- Whole group: tally 4 class surveys.
- Three stations.
- Stretch challenge at Station C: tally 10 classmates and total.
- Discuss the cross-out rule.
- Circulate to record formative evidence on the tally tracker.

3. Consolidation (13 mins): Tally's Counting Brag

Discuss:

- What is a TALLY?
- How do you draw 5?
- Why do we group by 5?
- What's the total of ||||CROSS ||||CROSS |||?

Exit: Each child draws ONE tally total: '____ tallies = ____'.

Lesson 3: Pictograph Power

Goal: Pictographs show data with PICTURES. One picture = one thing. Title and labels make it readable.

Expectation: D1.3

Learning Goal: "I can make a pictograph with pictures, a title, and labels."

Materials: M3_pictograph_grid, M4_picture_tokens, M7_data_chart, M10_detective_poster

1. Minds On (12 mins): Detective Dot Builds a Pictograph

Detective Dot: I asked '5 children: which fruit?' Now I show the data with PICTURES — one apple per child who picked apple! Detective Dot leads the class chant: 'ONE picture — ONE thing! Title, pictures, labels — case CLOSED!'

Bring children to the carpet. Detective Dot + Tally puppets. 'Friends — yesterday we tallied. Today we turn tallies into a PICTOGRAPH.'

2. Action (25 mins): Pictograph Stations

- Whole group: build a pictograph from a class survey.
- STRATEGY COMPARE (5 min): Display TWO ways to show 'tally of 5 apples' on the chart — (i) five same-size apples in a row, (ii) one large apple labelled '5'.
- Three stations.
- Stretch challenge at Station B: build a pictograph with 4 categories.
- Discuss the one-to-one rule.
- Circulate to record formative evidence on the pictograph tracker.

3. Consolidation (13 mins): Detective Dot's Picture Brag

Discuss:

- What is a PICTOGRAPH?
- What does ONE-TO-ONE mean?
- Where does the TITLE go?
- What do LABELS tell us?

Exit: Detective Dot call-and-response chant: teacher calls 'ONE picture —' and class chants back 'ONE thing!' twice.

Lesson 4: Order Greatest to Least

Goal: We can ORDER categories by HOW MANY. Greatest goes first, least goes last.

Expectation: D1.4

Learning Goal: "I can order categories from greatest to least."

Materials: M3_pictograph_grid, M4_picture_tokens, M7_data_chart, M10_detective_poster

1. Minds On (12 mins): Tally Orders the Pet Pictograph

Tally: Look at this pet pictograph. 4 dogs, 6 cats, 2 fish. Which has the MOST? Which has the LEAST?

Bring children to the carpet. Detective Dot + Tally puppets. 'Friends — yesterday we BUILT pictographs. Today we ORDER them.'

2. Action (25 mins): Ordering Stations

- Whole group: order 4 pictographs greatest to least.
- Three stations.
- Stretch challenge at Station C: order 5 categories (with one tie).
- Discuss tie-handling.
- Circulate to record formative evidence on the ordering tracker.

3. Consolidation (13 mins): Tally's Order Brag

Discuss:

- What does GREATEST mean?
- What does LEAST mean?
- What if two are tied?
- Why do detectives order data?

Exit: Each child shows ONE pictograph and names: 'Greatest = ____.

Lesson 5: The Detective's Conclusion Capstone

Goal: Detectives ASK questions of the data and write a CONCLUSION. The data tells the story.

Expectation: D1.5

Learning Goal: "I can ask and answer questions about a data display and write a conclusion."

Materials: M1_sort_objects, M2_tally_chart, M3_pictograph_grid, M4_picture_tokens

1. Minds On (12 mins): Detective Dot Solves a Case

Detective Dot: I have a class survey. ASK me a question. I'll answer with the DATA.

Bring children to the carpet. Detective Dot + Tally puppets in detective hats. 'Friends — today is our CAPSTONE. We use ALL 5 detective steps: ASK, SORT, TALLY, GRAPH, CONCLUDE.'

2. Action (25 mins): The Detective Case

- Each child gets a Detective Case Template.
- Step 1-2 (5 min): Ask + Survey.
- Step 3-4 (10 min): Tally + Graph.
- Step 5 (5 min): Conclude.
- Circulate to record summative evidence on the capstone rubric.

3. Consolidation (13 mins): Detective Dot's Case Brag

Discuss:

- What were the 5 detective steps?
- What is a CONCLUSION?
- How does the DATA tell the story?
- What was your favourite detective step?

Exit: Each child shares ONE conclusion: 'My data shows that ____.'

Name: _____

Date: _____

Detective Dot's First Case

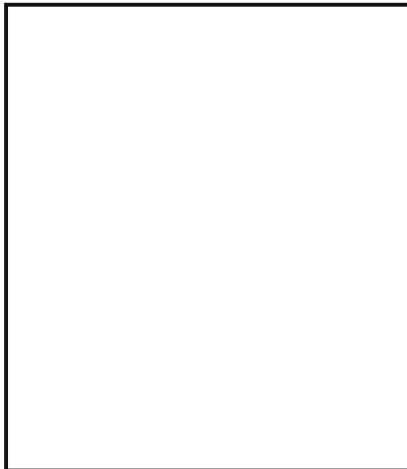
Learning Goal: I can sort things by one rule and say what the rule is.

Instructions for the Student:

1. Look at the pile.
2. Same pile as Part 1.
3. Pick a rule.

Sort by SHAPE

SQUARES



CIRCLES



Name: _____

Date: _____

Tally's Tally Time

Learning Goal: I can collect data and record it with tally marks.

Instructions for the Student:

1. Look at each tally.
2. Pick a question.
3. Add up these tallies.

My Tally Chart

Question: _____

Item	Tally	Total
Apples		
Bananas		
Grapes		

Name: _____

Date: _____

Detective Dot's Pictograph

Learning Goal: I can make a pictograph with pictures, a title, and labels.

Instructions for the Student:

1. Look at the pictograph.
2. The tally chart shows 4 categories.
3. Survey your family at home.

Make Your Own Pictograph

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--	--	--	--	--	--	--	--

--	--	--	--	--	--	--	--

Name: _____

Date: _____

Tally's Ordering Quest

Learning Goal: I can order categories from greatest to least.

Mission

1. Look at each pictograph.
2. Look at the tally chart.
3. Use this data: 4 hockey, 7 soccer, 3 baseball.



Name: _____

Date: _____

Detective Dot's Final Case

Learning Goal: I can ask and answer questions about a data display and write a conclusion.

Instructions for the Student:

1. Pick a question.
2. Use your tally to build a pictograph.
3. Look at your pictograph.

My Capstone Tally

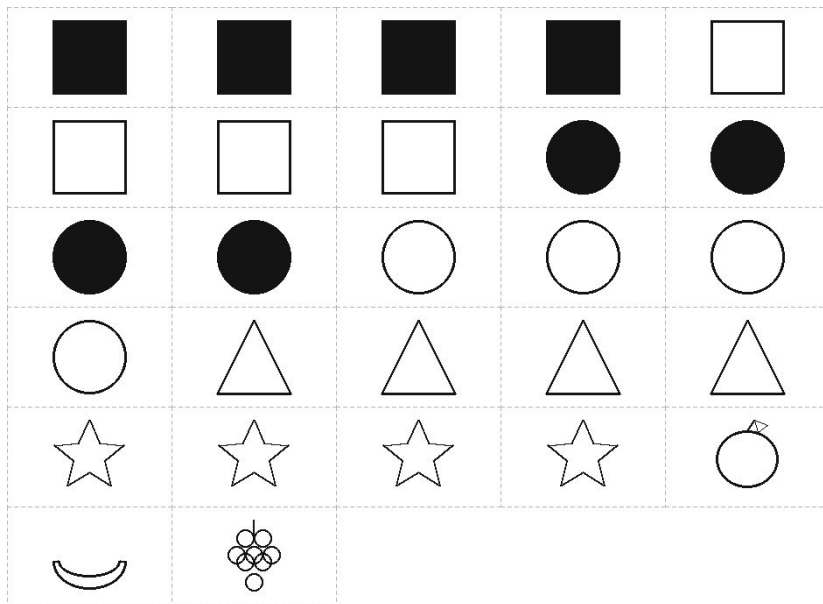
Question: _____

Item	Tally	Total

Sortable Object Cards

(M1_sort_objects)

A set of 30 paper object tokens for sorting by attributes (colour, shape, type) across Lessons 1 and 5.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 10 sets

Lessons: 1, 5

Prep: ~25 min

TEACHER PREP

1. Print 10 sets total (1 set per pair).
2. Laminate before cutting for repeated use.
3. Cut along the grid lines into 30 individual tokens per set.
4. Store each set in a labelled zip-top bag.

Tally Chart Template

(M2_tally_chart)

A letter-size template for recording tally marks with category labels.

My Tally Chart

Question: _____

Item	Tally	Total

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 10-12 templates

Lessons: 2, 5

Prep: ~10 min

TEACHER PREP

1. Print 12 templates (1 per pair plus 2 extras).

2. Laminate for dry-erase use across lessons 2 and 5.

3. Distribute one template per pair.

Pictograph Grid Mat

(M3_pictograph_grid)

A letter-size grid mat for building pictographs with one-to-one correspondence in Lessons 3, 4, and 5.

My Pictograph

PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS

Quantity: 10-12 mats
Lessons: 3, 4, 5
Prep: ~10 min

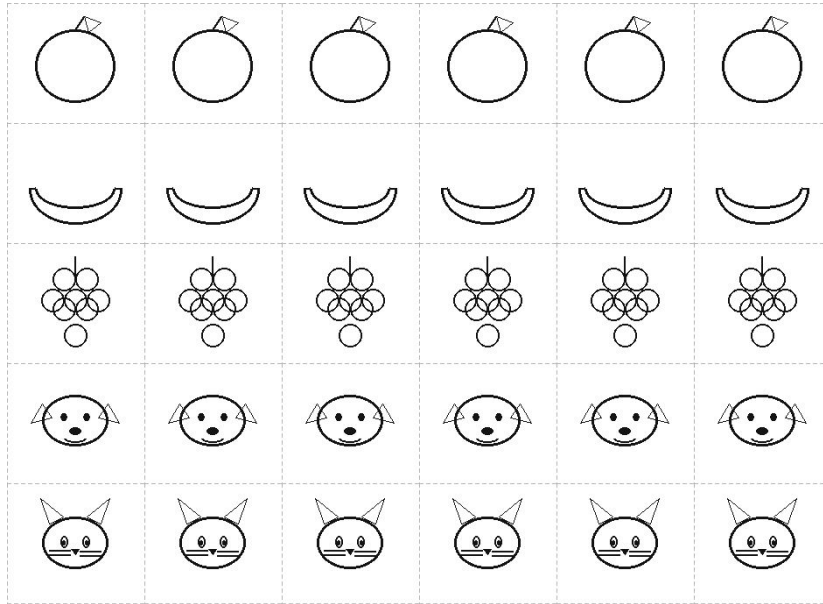
TEACHER PREP

1. Print 12 mats (1 per pair plus 2 extras).
2. Laminate for dry-erase use across lessons 3, 4, 5.
3. Distribute one mat per pair.

Pictograph Picture Tokens

(M4_picture_tokens)

A set of 60 small pictorial tokens (fruits, pets, sports) for placing on the pictograph grid with one-to-one correspondence.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 10 sets

Lessons: 3, 4, 5

Prep: ~30 min

TEACHER PREP

1. Print 10 sets (1 per pair).
2. Laminate before cutting for repeated use.
3. Cut along the grid lines into 60 individual tokens per set.
4. Store each set in a labelled zip-top bag.

Survey Question Cards

(M5_question_cards)

A set of 12 pre-written survey question cards for collecting data.

Do you have a pet? ☐ ☐ ☐	Favourite fruit? ☐ ☐ ☐	Favourite colour? ☐ ☐ ☐
Cats or dogs? ☐ ☐ ☐	Do you walk to school? ☐ ☐ ☐	Favourite season? ☐ ☐ ☐
Favourite drink? ☐ ☐ ☐	Favourite snack? ☐ ☐ ☐	Do you read at night? ☐ ☐ ☐
Favourite sport? ☐ ☐ ☐	Best subject? ☐ ☐ ☐	Hot or cold drink? ☐ ☐ ☐

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 10 sets

Lessons: 2, 5

Prep: ~15 min

TEACHER PREP

1. Print 10 sets (1 per pair).
2. Laminate before cutting.
3. Cut along the grid lines into 12 individual question cards per set.
4. Store each set in a labelled zip-top bag.

Detective Case Template

(M6_capstone_template)

A letter-size template for the Lesson 5 capstone with sort area, tally chart, pictograph, ordering, and conclusion line.

1. My QUESTION:
2. My DATA (tally):
3. My PICTOGRAPH:
4. What I FOUND:
5. My CONCLUSION:

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

USE IN CLASS

Quantity: 25 templates

Lessons: 5

Prep: ~5 min

TEACHER PREP

1. Print 25 templates (1 per child plus 5 extras).

2. Stack near the teacher table for distribution at the start of Lesson 5 Action.

3. No lamination needed — these are single-use summative artifacts.

Data Vocabulary Chart

(M7_data_chart)

A teacher-display anchor chart of data vocabulary (sort, tally, pictograph, title, label, conclusion).

Data Words	
DATA	Information we collect to answer a question.
TALLY	Marks we use to count. Five marks = a bundle.
PICTOGRAPH	A graph that uses pictures to show data.
CATEGORY	A group that things belong to.
MOST	The category with the largest number.
LEAST	The category with the smallest number.

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 chart

Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print 1 copy on standard letter paper.

2. Laminate for repeated reference across the unit.

3. Tape to the front classroom wall before Lesson 1.

Tally Marks Reference

(M8_tally_chart)

A teacher-display anchor chart showing 1, 2, 3, 4, 5 (with cross-out) tally mark conventions.

Tally Marks Reference

1	2	3	4	5
				
marks	marks	marks	marks	marks

PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy on standard letter paper.

2. Laminate for repeated reference.

3. Tape next to the Data Vocabulary Chart.

USE IN CLASS

Quantity: 1 chart

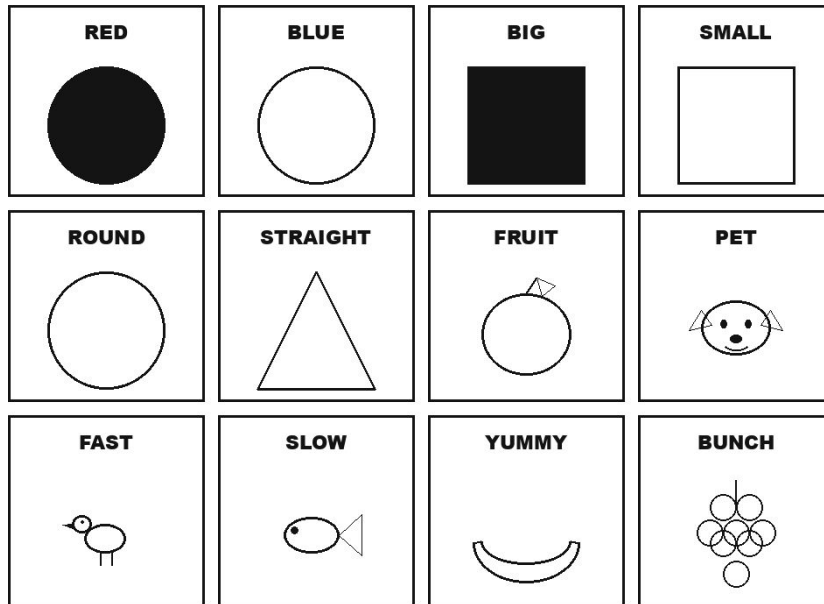
Lessons: 2, 5

Prep: ~10 min

Attribute Cards

(M9_attribute_cards)

A set of 12 attribute cards (red, blue, big, small, round, square, etc.)
for sorting rules.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 10 sets

Lessons: 1, 5

Prep: ~15 min

TEACHER PREP

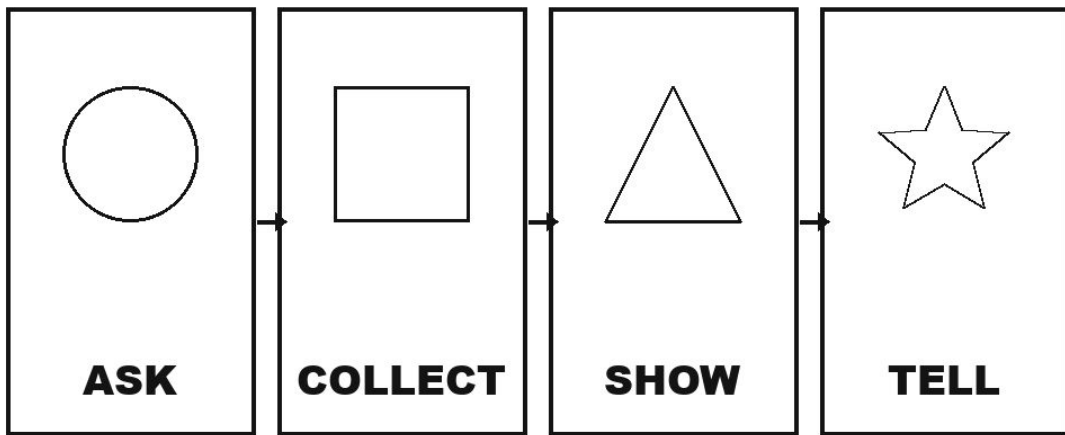
1. Print 10 sets (1 per pair).
2. Laminate before cutting.
3. Cut along the grid lines into 12 individual cards per set.
4. Store each set in a labelled zip-top bag.

Data Detectives Poster

(M10_detective_poster)

A unit anchor poster of Detective Dot and Tally with the 5 detective steps: ASK, SORT, TALLY, GRAPH, CONCLUDE.

Data Detectives!



Ask. Collect. Show. Tell.

PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 poster

Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print 1 copy on letter or 11x17 paper.

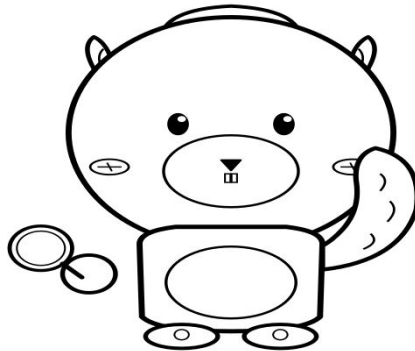
2. Laminate the poster for the full unit duration.

3. Tape to the front classroom wall on Day 1 before Lesson 1 begins.

Detective Dot Puppet

(char_dot_puppet)

A paper puppet of Detective Dot the Sorting Squirrel used by the teacher in the Minds-On phase of every lesson.



Detective Dot the Squirrel

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 1, 2, 3, 4, 5

Prep: ~12 min

TEACHER PREP

1. Print 1 copy of Detective Dot on cardstock if possible.

2. Cut around the outside of Dot's silhouette including the bottom tab.

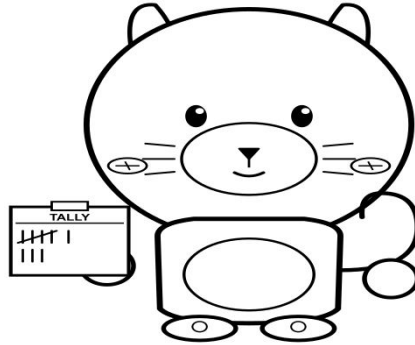
3. Laminate before assembly for durability.

4. Glue or tape a popsicle stick behind the bottom tab so Dot can be held up.

Tally Puppet

(char_tally_puppet)

A paper puppet of Tally the Tally Cat used by the teacher in Lessons 2-5 to demonstrate counting.



Tally the Tally Cat

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 2, 3, 4, 5

Prep: ~12 min

TEACHER PREP

1. Print 1 copy of Tally on cardstock if possible.

2. Cut around the outside of Tally's silhouette including the bottom tab.

3. Laminate before assembly for durability.

4. Glue or tape a popsicle stick behind the bottom tab so Tally can be held up.

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
D1.1 sort sets of data about people or things...	Cannot consistently apply one rule.	Sorts by one rule with adult prompting.	Independently picks a rule, applies it consistently across the set, and names...	Re-sorts the same set by a different rule and explains how the rule changes...
D1.2 collect data through observations, experiments,...	Cannot independently ask a survey question.	Asks a question with a frame.	Independently asks a question, surveys 5+ classmates, draws 4+1 tallies, and...	Reads tally, totals by 5s + ones, AND chooses fast strategy from 4+1 cross...
D1.3 display sets of data, using one-to-one...	Draws one big picture for many items.	Draws one-to-one with prompts.	Builds pictograph with one-to-one same-size pictures, title and labels —...	Builds pictograph one-to-one with title + labels AND explains why size =...
D1.4 order categories of data from greatest to least...	Orders by row position rather than count.	Identifies greatest and least with prompts.	Independently identifies greatest and least, then orders all categories...	Handles ties correctly and orders 5+ categories.
D1.5 analyse different sets of data presented in...	Writes a personal preference instead of a data-based conclusion ('I like...').	Writes a group conclusion with sentence frame help ('Most children chose ____').	Independently writes a clear group conclusion using 'most' or 'least' that...	Owens full 5-step case AND shows another detective how each step holds up the...

Junior Data Detective Certificate

This certificate is proudly presented to:

For successfully completing the 5-Day Data Detectives unit and demonstrating the skills of a Junior Data Detective. A data detective certificate. Detective Dot the Sorting Squirrel congratulates the student.

By completing this unit, this student has demonstrated:

- Sorting things by a single rule
- Collecting data with tally marks
- Building pictographs with title and labels
- Ordering categories greatest to least and writing conclusions

Marketplace Listing

A 5-day Grade 1 unit on data framed around Detective Dot the Sorting Squirrel and Tally the Tally Cat.

Price: \$8.99 CAD

Pages: ~45

Classroom time: 250 min

Teacher prep: ~200 min

Top tags: Grade 1, math, data, data management, tally marks, pictograph, survey, sorting

What's included:

- Unit blueprint with curriculum mapping (1 file)
- 5 lesson plans with minds-on, action, consolidation, expanded walkthroughs (5 files)
 - 5 student worksheets with answer keys and image placeholders (5 files)
 - Manipulatives plan with 12 print-ready assets (1 file: sortable objects, tally chart, pictograph grid, picture tokens, survey cards, capstone template, anchor...
 - Formative + reflection sheets (1 file)
 - Assessment suite (1 file: diagnostic tracker, 3 formative trackers, summative rubric, summative administration guide, Junior Data Detective certificate)